Pre-reconstruction RSD P0 + B000 | contract and interpretation

Status: COMPLETE | summaries 18/18 | converged 18/18

P0: same real-space 12-bin edges/density, kmax_P = 0.08 h/Mpc; actual CPU window.

B000: two measured legs, same six cuts 0.08 / 0.10 / 0.12 / 0.14 / 0.18 / 0.20.

Targets: same 100 RSD boxes per truth 0 / +100 / -100; covariance: independent Fid500.

Full matched P-B cross covariance retained. C_single is NOT divided by 100 or 500.

Seven B-only parameters/priors/model unchanged; one independent residual Pshot added.

Sampling: original four independent ensembles x 32 walkers; identical convergence gates.

Interpretation: a single-box-precision model test, not the likelihood of the mean100.

Wide credible intervals covering the truth are not proof of an unbiased physical model.

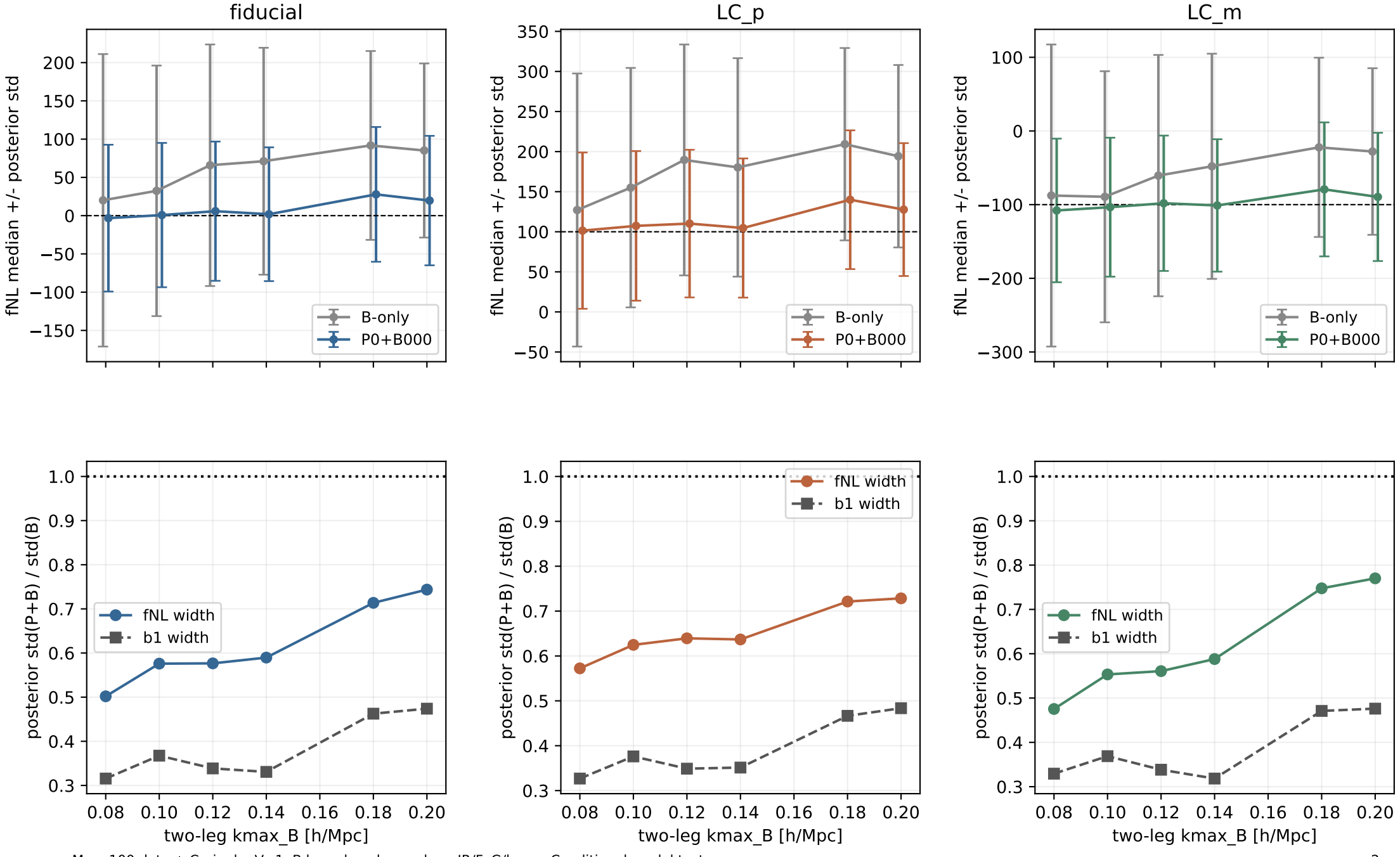
P0 can break bias/PNG/noise degeneracy; it cannot repair missing nonlinear velocity physics.

No IR, FoG, loops, counterterms, or higher multipoles included in this controlled comparison.

Finite-PNG B stochastic closure and PBS recipe are unchanged, explicitly provisional.

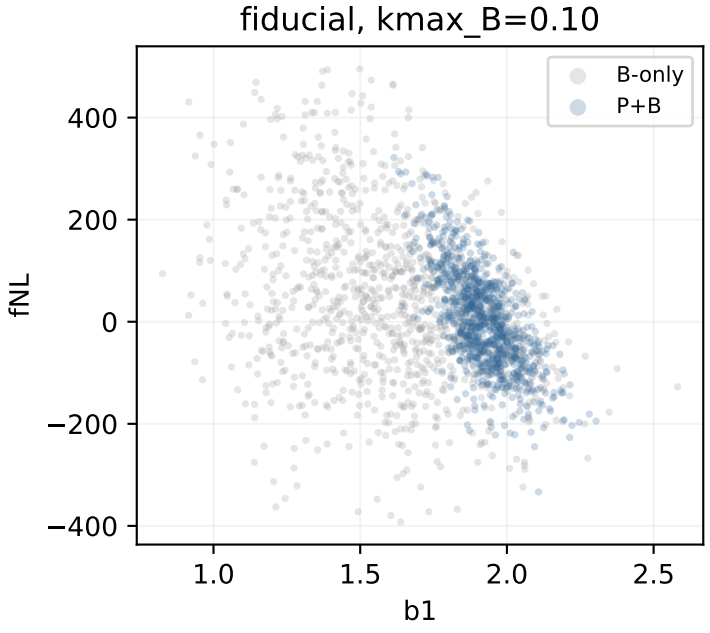
Numerical mesh and frozen-input checks precede production, independently of truth recovery.

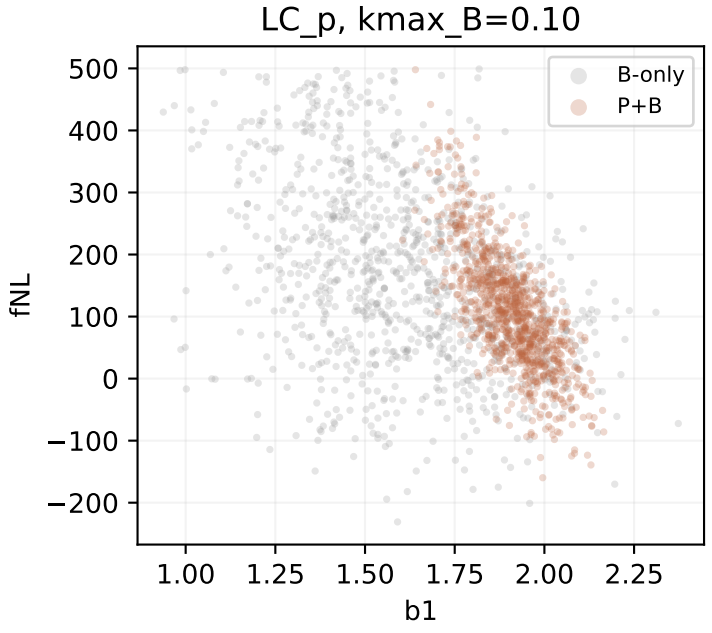
Pre-reconstruction RSD P0 + B000 | fNL recovery and width ratios

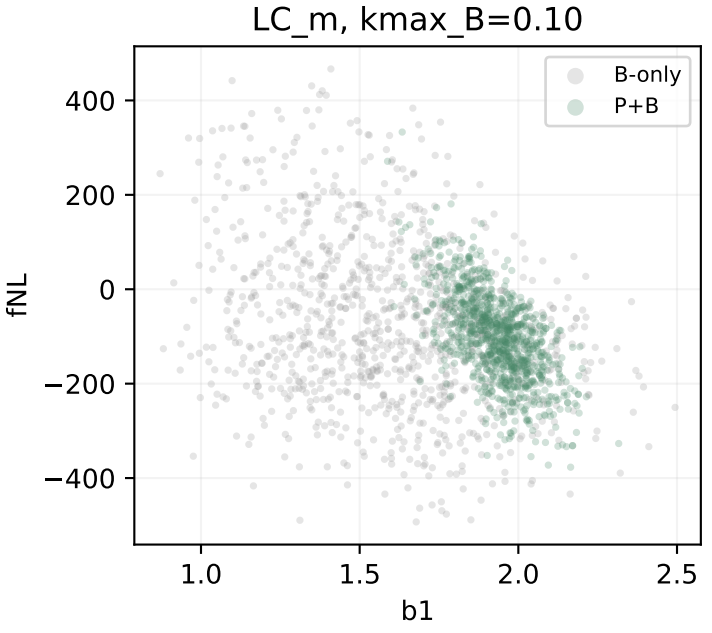


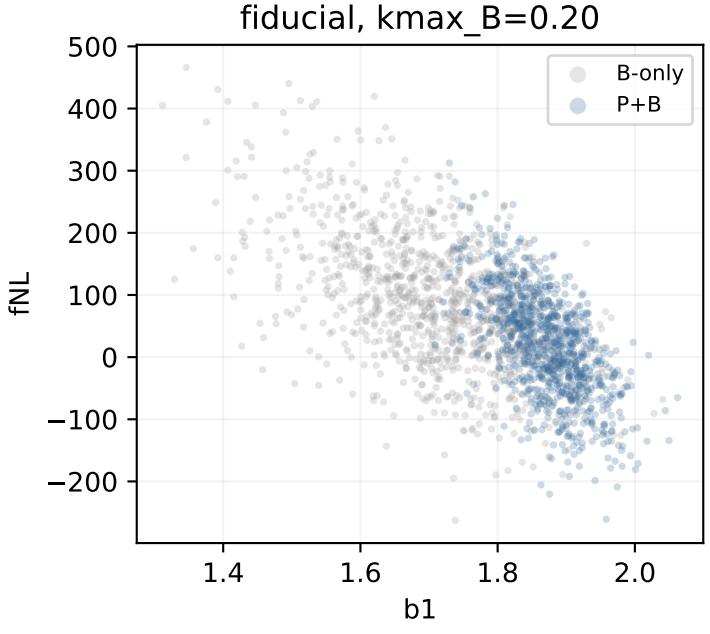
Mean100 data + C_single; V=1. B kernel unchanged; no IR/FoG/loops. Conditional model test.

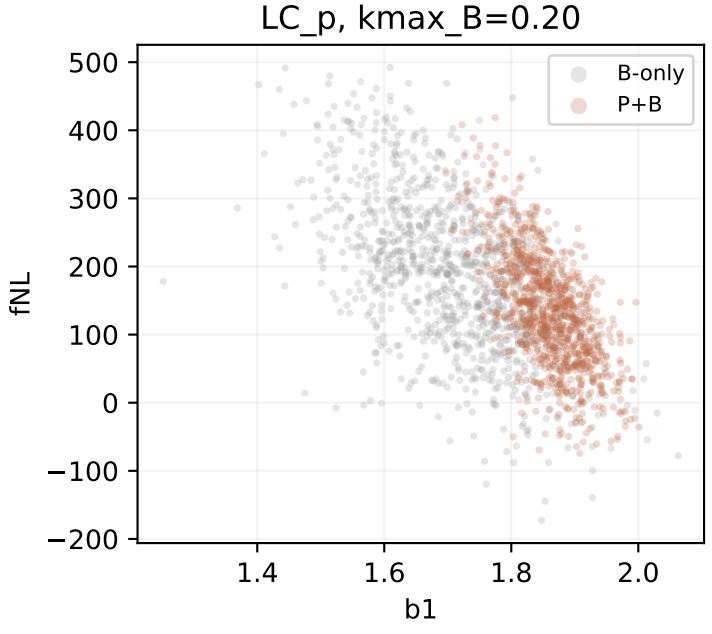
Pre-reconstruction RSD P0 + B000 | bias-PNG degeneracy

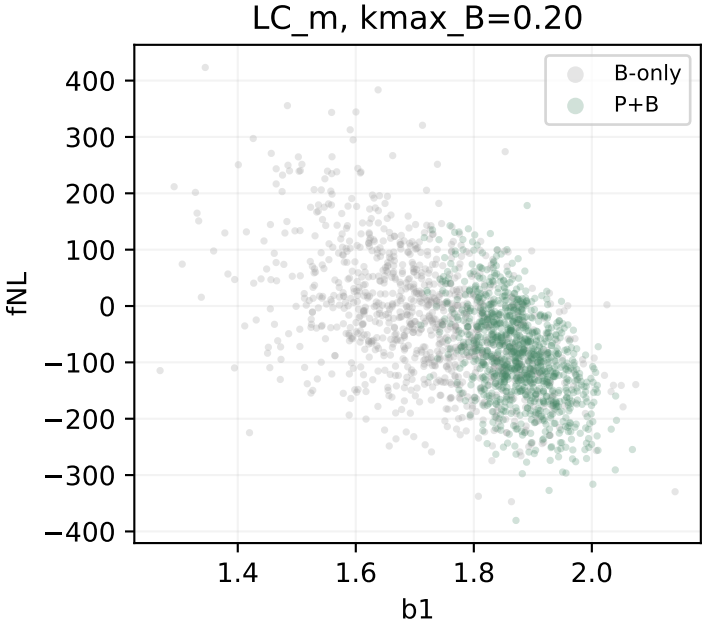






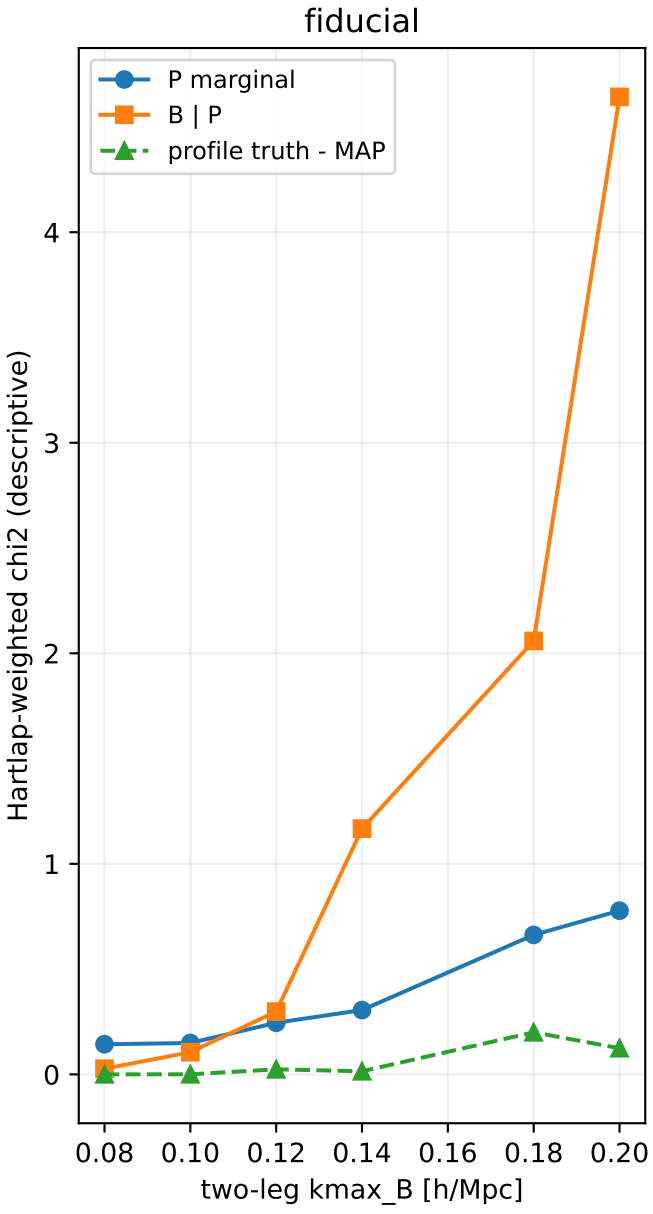


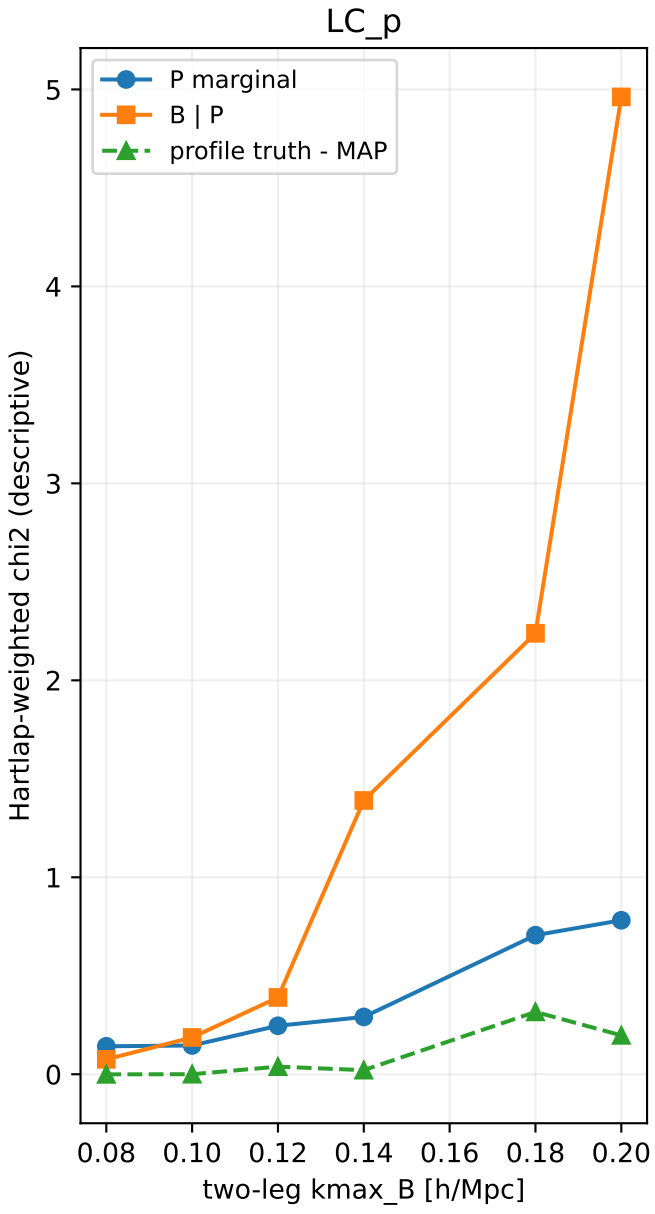


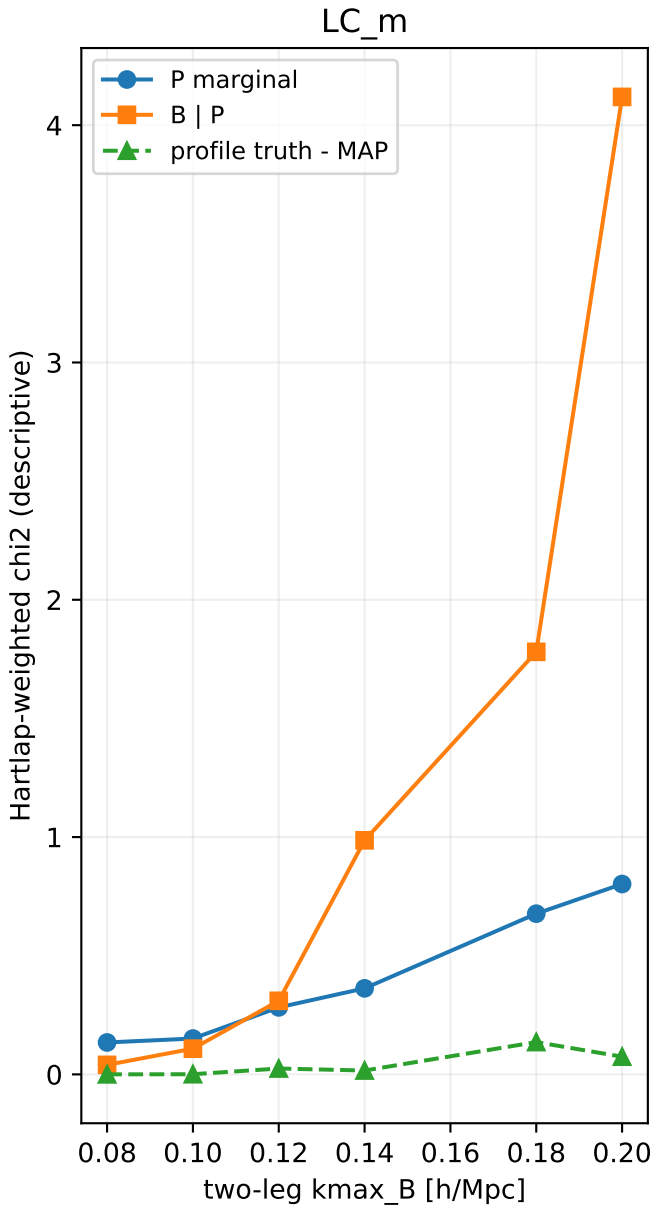


Mean100 data + C_single; V=1. B kernel unchanged; no IR/FoG/loops. Conditional model test.

Pre-reconstruction RSD P0 + B000 | full-covariance conditional residuals

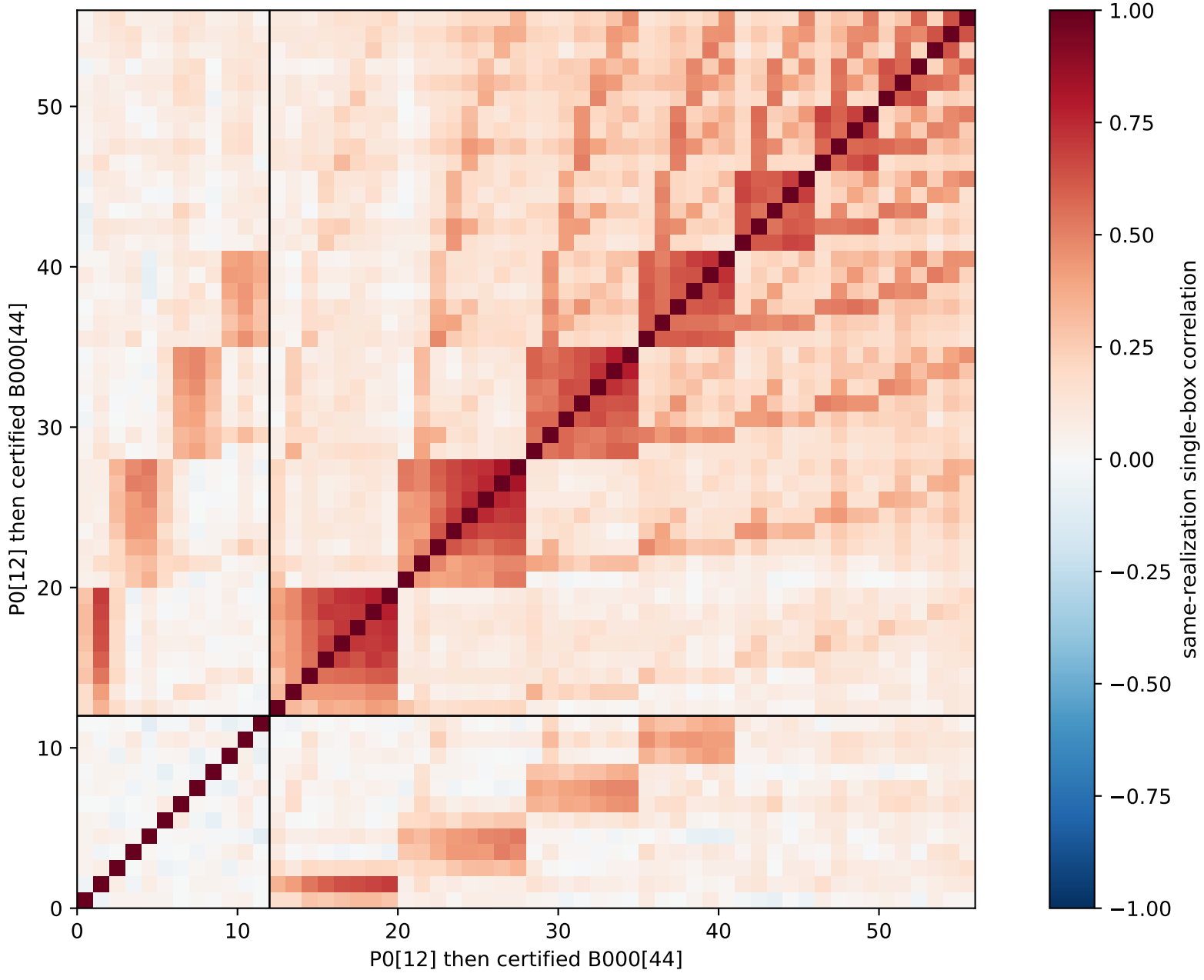






Mean100 data + C_single; V=1. B kernel unchanged; no IR/FoG/loops. Conditional model test.

Pre-reconstruction RSD P0 + B000 | matched joint covariance



Mean100 data + C_single; V=1. B kernel unchanged; no IR/FoG/loops. Conditional model test.

Pre-reconstruction RSD P0 + B000 | all joint MCMC summaries

sample	kB	fNL	std	pull	b1	std ratio	Rhat	steps
fiducial	0.08	-3.26	95.94	-0.03	1.931	0.502	1.00076	12000
fiducial	0.10	0.76	94.30	0.01	1.917	0.576	1.00097	10000
fiducial	0.12	5.83	91.01	0.06	1.915	0.577	1.00131	10000
fiducial	0.14	1.85	87.44	0.02	1.916	0.590	1.00069	8000
fiducial	0.18	27.80	88.07	0.32	1.872	0.714	1.00074	8000
fiducial	0.20	19.69	84.61	0.23	1.869	0.744	1.00082	8000
LC_p	0.08	101.42	97.50	0.01	1.918	0.572	1.00049	10000
LC_p	0.10	107.35	93.31	0.08	1.906	0.625	1.00105	10000
LC_p	0.12	110.20	92.10	0.11	1.905	0.639	1.00052	8000
LC_p	0.14	104.68	86.79	0.05	1.907	0.637	1.00046	10000
LC_p	0.18	139.96	86.54	0.46	1.857	0.721	1.00083	8000
LC_p	0.20	127.67	82.89	0.33	1.861	0.728	1.00102	8000
LC_m	0.08	-107.89	97.46	-0.08	1.946	0.475	1.00074	12000
LC_m	0.10	-103.43	94.30	-0.04	1.928	0.553	1.00100	10000
LC_m	0.12	-98.13	91.85	0.02	1.919	0.561	1.00130	8000
LC_m	0.14	-101.09	89.86	-0.01	1.917	0.588	1.00084	8000
LC_m	0.18	-79.29	90.95	0.23	1.880	0.748	1.00111	8000
LC_m	0.20	-89.51	87.07	0.12	1.875	0.770	1.00092	8000